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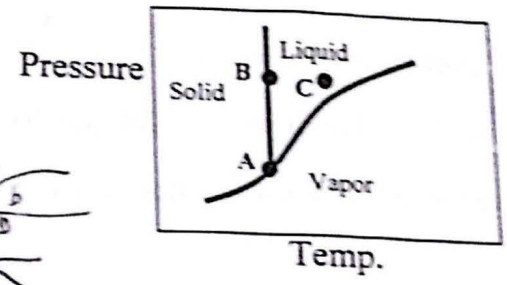
Last, First Name: ___ _____**Student ID:** _____**Sec:**

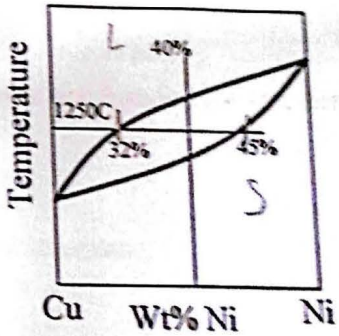
1. Gibbs rule describes the state of a material and has the general form of F=C-P+2. If the phase diagram of a pure metal given as shown below, which of the following statements would present three points A,B and C?



(a) @A: F=0, @ B: F=-1, @ C: F=2
(b) @ A: F=0, @ B: F=1, @ C: F=1
(c) @ A: F=1, @ B: F=0, @ C: F=2
(d) @ A: F=0, @ B: F=1, @ C: F=2

2. A system is at equilibrium if
(a) All phases of the system consist of the same crystal structure.
(b) All phases of the system consist of the same concentrations at a constant temperature.
(c) The system includes the minimum randomness or disorder of the atoms.
(d) The system possesses the highest energy and thermal energy.

3. Binary phase diagram (Ni-40% Cu) is shown.
The tie line at 1250°C represents:
(a) Solid phase is first nucleated when it contains 32% Ni.
(b) Liquid contains 32% Ni and solid contains 45% Ni.
(c) Solid contains 32% Ni and liquid contains 45% Ni.
(d) Both (a) and (b)
(e) Both (a) and (c)



4. Non-equilibrium solidification results in the formation of a multi-layer solid phase with different layer concentrations. The non-equilibrium condition is due to:
(a) A low rate of solidification which weakens the strength of materials.
(b) A high rate of solidification which weakens the strength of materials.
(c) A low rate of solidification which improves the strength of materials.
(d) A high rate of solidification which improves the strength of materials.

5. Which one of the following answers represents reactions (1) $\eta+L\rightarrow\alpha$, (2) $L\rightarrow\alpha+\beta$, (3) $L1\rightarrow\beta+L2$, and (4) $\alpha\rightarrow\delta+\beta$, respectively.
(a) (1) Peritectic, (2) Eutectoid, (3) Monotectic, and (4) Eutectic
(b) (1) Monotectic, (2) Eutectic, (3) Peritectic, and (4) Eutectoid
(c) (1) Peritectic, (2) Eutectic, (3) Monotectic, and (4) Eutectoid
(d) (1) Peritectic, (2) Eutectic, (3) Eutectoid, and (4) Monotectic

6. In a Peritectic reaction the phase transformation mechanism is based on:
(a) The nucleation of α and β solid phases during solidification of the liquid phase L.
(b) The formation of the second solid phase β as a consequence of rapid quenching of liquid phase L.

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(c) The formation of a new solid phase β as a consequence of diffusing atoms of liquid phase L and solid phase α.
(d) All the above
(e) None of the above.

7. In alloying systems. *Solubility limit* can be defined as
(a) The maximum concentration of solute atoms that may solve in the solvent to form solid solution.
(b) The minimum concentration of solute atoms that may solve in the solvent to form solid solution.
(c) The maximum concentration of solvent atoms that may solve in the solute to form solid solution.
(d) All of the above.

8. Crystal structures and properties of ceramic materials are generally influenced by:
(a) The magnitude of electrical charges of Cations and Anions.
(b) The relative size of Cations and Anions.
(c) The number of Anions surrounding the central Cation.
(d) Both (a) and (b)
(e) Both (b) and (c)

9. The graphite structure is composed of: Parallel hexagonal layers are bonded by van der Waals bonds.
(a) Both Pentagon and Hexagon Carbon layers.
(b) Hexagonal layers of Carbon atoms bonded covalently.
(c) Covalently bonded sixty atoms of Carbon to form a spherical structure.
(d) Covalently bonded four atoms of Oxygen to a Silicon atom.

10. In ionically bonded ceramics, (1) Cations and (2) Anions are respectively defined as:
(a) (1) Metallic ions and positively charged (2) non-metallic ions and negatively charged.
(b)(1) Non-metallic ions and negatively charged (2) Metallic ions and positively charged.
(c) (1) Both Cation and Anions are non-metallic ions.
(d) None of the above.

11. Critical radius ratio of Cation-Anion represents:
(a) That the solid solution of a Cation and surrounding Anions are strong.
(b) That the structure of ceramic materials is stable.
(c) The coordination number of Cation-Anion.
(d) Both (a) and (c)
(e) Both (b) and (c)

12. In polymers, the relaxation modulus and rigidity vary with temperature change. Which one of the following answers presents the status of the polymer respectively for: poor ductility, low strength, and viscous states:
(a) Rubbery, Glassy and Liquid.
(b) Rubbery, Liquid and Glassy.

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(c) Glassy, Rubbery and Liquid.
(d) Glassy, Liquid and Rubbery.
(e) Liquid, Glassy and Rubbery.

13. Low Density (LD) polyethylene has a tensile strength which is about half of tensile strength of High Density (HD) polyethylene because
(a) LD polyethylene has a more number of branches than HD polyethylene.
(b) LD polyethylene has a lower number of branches than HD polyethylene.
(c) LD and HD polyethylene have anisotropic behavior.
(d) Both (a) and (c).
(e) None of the above.

14. The pure non-linear elastic deformation on Elastomers is due to:
(a) Addition mechanism of polymerization in Elastomers.
(b) A complex deformation of polymeric chains due to side branches.
(c) Cross-linking of the polymeric chains.
(d) Both (a) and (c)
(e) Both (b) and (c)

15. In polymerization process, Addition mechanism differs from Condensation mechanism because
(a) In addition mechanism, there are different types of monomers while in Condensation mechanism, monomers are identical.
(b) In Addition and Condensation mechanisms, there are different types of reaction initiators and terminators involved.
(c) In Condensation mechanism, there are different types of monomers while in Addition mechanism, monomers are identical.
(d) In Condensation mechanism, a higher activation energy is required as the Addition mechanism requires a lower energy of reaction.
(e) None of the above.

16. Which one of the following choices characterizes the given polymer by its property
(a) Lucite is transparent
(b) Nylon possesses an electrical conductivity
(c) Teflon has a low coefficient of friction
(d) All of the above
(e) Both (a) and (c)

17. Which one of the following answers represents variables affecting the glass transition temperature?
(a) The higher free volume of polymer, the lower the T_g.
(b) The higher attractive forces between molecules of chains, the higher the T_g.
(c) The lower the chain mobility, the higher the T_g.
(d) The longer the chain length, the higher the T_g.
(e) All of the above

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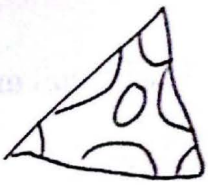
18. Which one of the following statements corresponds respectively to (i) Thermoplastic, (ii) Thermosetting and (iii) Elastomer polymers?
(a) (i) Epoxies, Melamine Formaldehyde, (ii) Polyethylen, Nylon, (iii) Natural Rubber, Neoprene.
(b) (i) Polyethylen, Nylon, (ii)Epoxies, Melamine Formaldehyde, (iii) Natural Rubber, Neoprene.
(c) (i) Natural Rubber, Neoprene (ii)Epoxies, Melamine Formaldehyde, (iii) Polyethylen, Nylon.
(d) (i) Polyethylen, Nylon, (ii) Natural Rubber, Neoprene (iii) Epoxies, Melamine Formaldehyde.

19. Plastic deformation in polymers is
(a) Due to dislocation motion along the polymeric chains.
(b) Due to crystallographic motion along the polymeric chains.
(c) Due to rotation and slip of molecular chains.
(d) Due to both crystalline dislocation motion and slip of molecular chains.

20. In CaO with an FCC crystal structure, the total number of anions on the plane (111) is: (Ionic radii: Ca^{2+}=0.106-nm and O^{2-}=0.132-nm)
(a) 1.0 atom area
(b) 1.5 atom areas
(c) 2.0 atom areas
(d) 2.5 atom areas
(e) 3.0 atom areas

21. To achieve a critical radius ratio of Cation-Anion in a ceramic structure with coordination number of 6 (CN=6) and with Anion of radius R=0.181nm and Cation of radius r, the Cation radius should be:
(a) r≥0.06 nm
(b) r≥0.150 nm
(c) r≥0.075 nm
(d) r≥0.120 nm
(e) r≥0.140 nm

$= \frac{C}{A}$ 0.181



~~1 0.33~~
~~5 0.828~~
~~2 0.414~~
7 0.66
4 0.77

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26. The degree of polymerization for a Benzoyl Peroxide($C_{14}H_{10}O_4$) with the molecular weight of 150,000 g/mol, is: (where C=12, H=1, and O=16 g/mol)
(a) 6198 mers/mol
(b) 620 mers/mol
(c) 5357 mers/mol
(d) 536 mers/mol

242g/mol

27. The average molecular weight, M_m={Sum(f_iM_i)/ Sum (f_i)}, for a thermoplastic material with following molecular weight fraction f_i for the molecular weight ranges listed below is:

Molecular wt Ranges (g/mol)	M_i	f_i
15,000-20,000	17,500	0.31
20,000-35,000	27,500	0.20

$M_m = \frac{5425 + 5500}{0.51}$

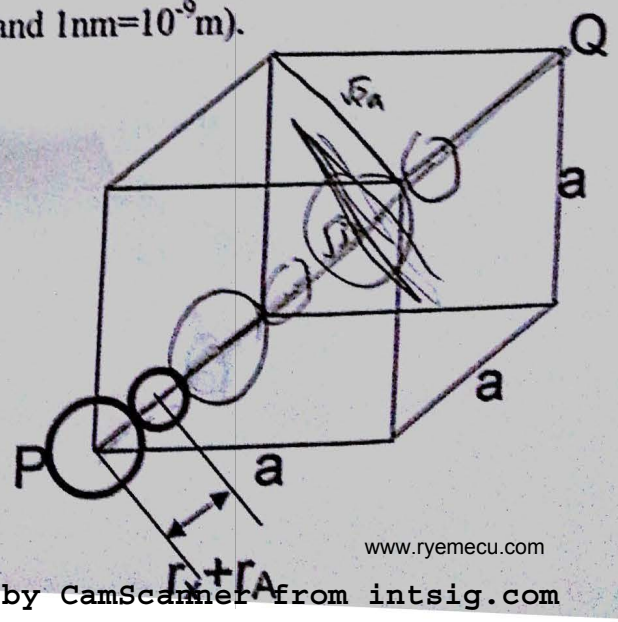
(a) 19,404
(b) 25,200
(c) 18,320
(d) 21,421

28. In the polymerization process of Ethylene (C_2H_4), Hydrogen peroxide (H_2O_2) molecule acts as the initiator and terminator of polyethylene chains. Hydrogen, Oxygen and Nitrogen possess atomic weights of 1, 16, and 12 g/mol respectively.
The number of C_2H_4 molecules and the number of chains involved in polymerization of 10,000g/mol of polyethylene respectively is: C_2H_4 = 28g/mol H_2O_2 = 34g/mol
(Note: $N_A=6.02\times10^{23}$)
(a) 28 and 34 2.15×10^{26}
(b) 357 and 294
(c) 1.7×10^{-22} and 2.15×10^{-26}
(d) 2.15×10^{26} and 1.7×10^{22}
(e) 2.15×10^{-26} and 1.7×10^{-22}

29. AX_2 ceramic structure consists of A^{+4} and X^{-2} ions. The ionic radius of A^{+4} =0.105 nm and that of X^{-2} =0.132 nm. The atomic mass of A=238 g/mol and that of X=16 g/mol. Assuming that there are four pairs of (r_A+ r_X) on PQ corner-to-corner distance, the volumetric density of AX_2 is calculated to be:
(Note: there are 4 A ions and 8 X ions. ρ=m/V and 1nm=10^{-9}m).
($N_A=6.02\times10^{23}$)

$\sqrt{3}a = (4(r_x+r_A))$
$a = 0.5473$ nm
$m = 4(238) + 8(16)$
$m = 1080$

(a) ρ=4.12 g/cm^3
(b) ρ=6.12 g/cm^3
(c) ρ=8.12 g/cm^3
(d) ρ=10.96 g/cm^3



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